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उपसमूहको प्रतियोगितात्मक लिखित परीक्षा
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विषय: बिल्डिंग टेक्नोलोजी सम्बन्धी |

1. **a. What are the sources of dampness in buildings and its negative effects? Sketch the section to internal waterproofing of basement using tar felt.**

Answer:

The presence or penetration of moisture in a building through walls, floors, roofs, or foundations is known as dampness. It affects the durability, appearance, and safety of a structure.

The sources of dampness in buildings are listed below:

i. Rain penetration:
   - Rainwater enters through cracks, porous walls, defective plaster and poorly sealed joints and creates dampness in the buildings.

ii. Roof leakage:
   - It is caused by damaged roofing materials, faulty materials and poor roof slopes.

iii. Groundwater leakage:
   - Water enters the basements and walls due to hydrostatic pressure and causes dampness.

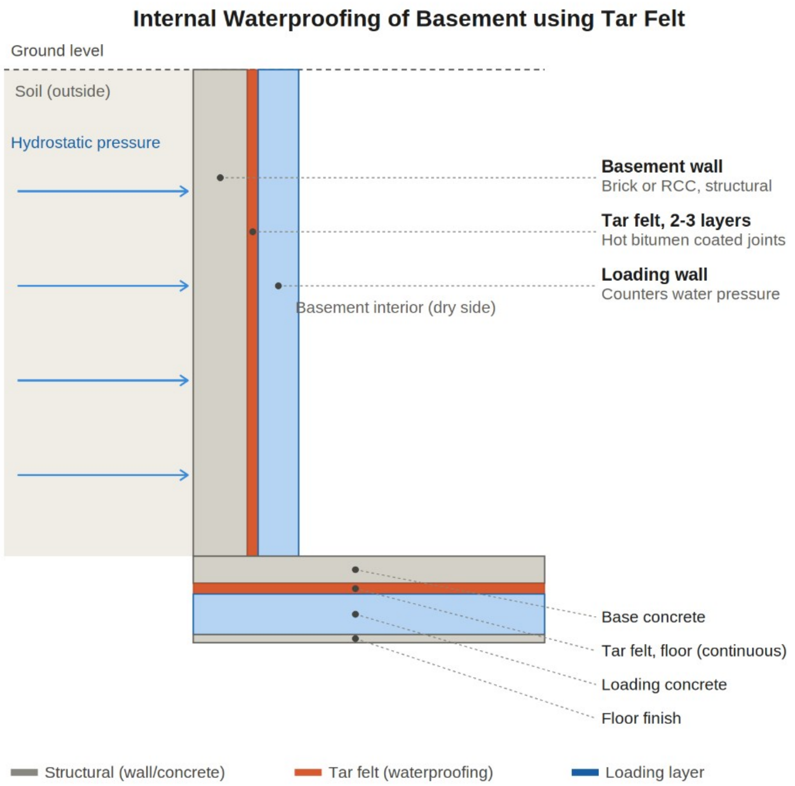
iv. Condensation:
   - Moisture forms on cold surface when warm and humid air comes in contact with them.

v. Defective drainage:
   - Poor site drainage causes water to accumulate around foundation of building which causes dampness.

vi. cracks in walls and floors:
   - structural cracks allow moisture to penetrate through cracks and floors.

The negative effects of dampness are stated below:

i. Dampness weakens the strength and durability of building materials.
ii. It causes damage in plaster and cause efflorescence.
iii. Dampness promotes growth of molds, algae and fungi.
iv. It produces unhealthy indoor conditions leading to respiratory problems.
v. It damages electrical wirings and fittings.
vi. It increases maintenance and repair costs of the buildings.

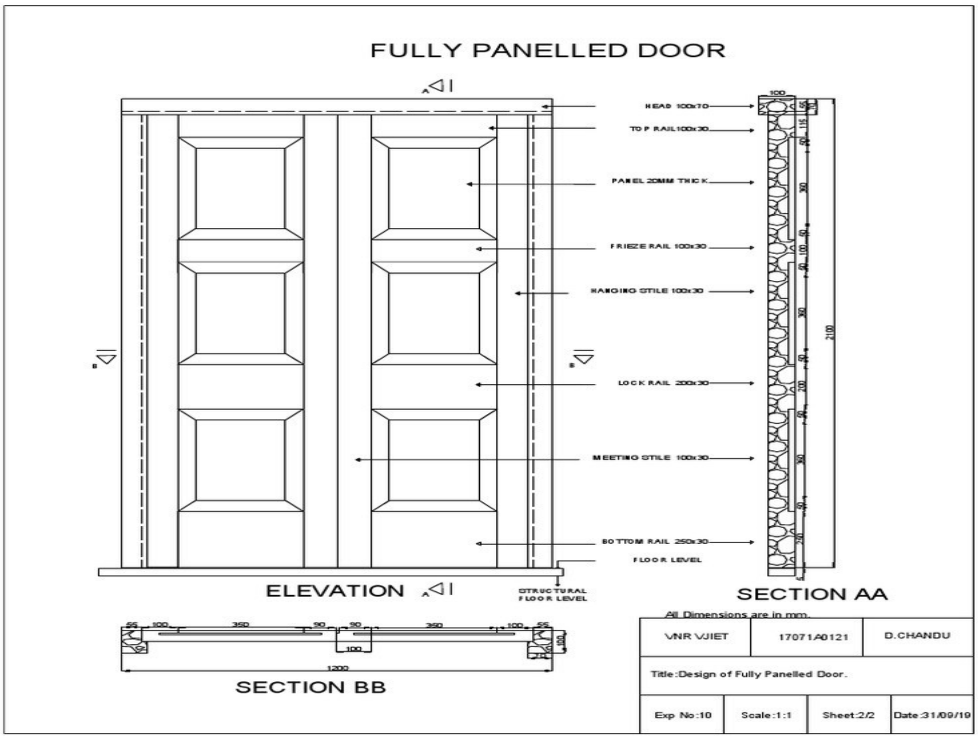


2. **Describe with neat sketch of the following:**
 a. Framed and paneled doors b. Sky light

Answer:

a. framed and paneled doors:

It is a door in which a framework of vertical and horizontal timber members forms the main structure, while the spaces between them are filled with wooden, plywood, glass, or other panels. It is most commonly used doors in residential and institutional buildings because of its strength, durability, and attractive appearance.



The main components of framed and paneled doors are as follows:

i. Stiles:
   - It is a vertical member forming the two sides of door.

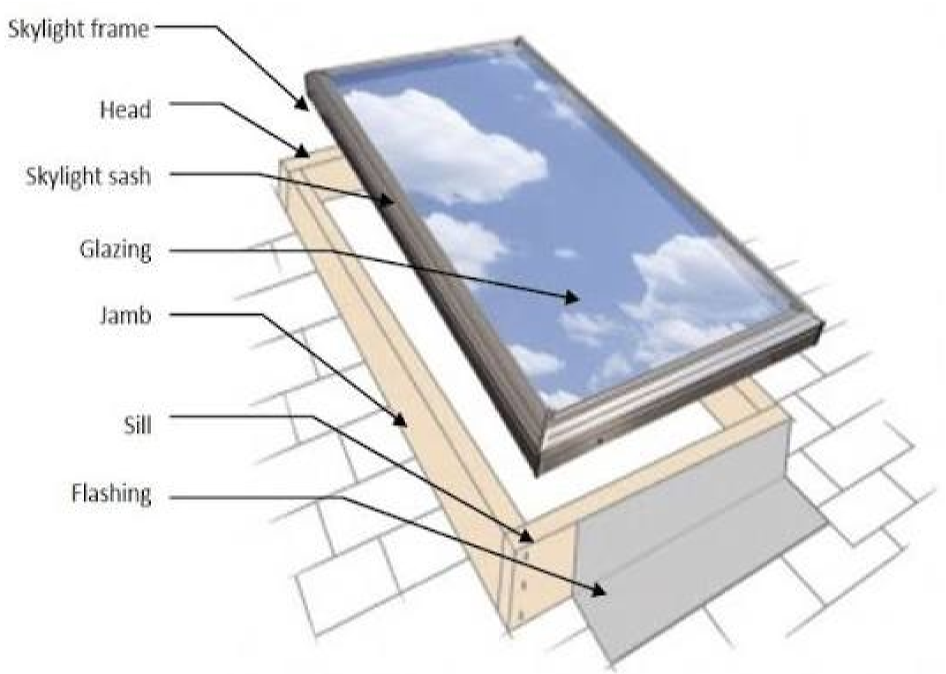
ii. Rails:
   - It is a horizontal member connecting the stiles, which consists of top rail, local rail and bottom rail.

iii. Panels:
   - The infill portions are fitted within the frame.
   - They are made of timber, plywood, glass etc.

iv. Mutins:
   - It is an optional intermediate vertical member dividing large panels into smaller ones.

b. Sky light:

A **s**kylight is a glazed opening or window provided in the roof or sloping roof of a building to admit natural daylight and, in some cases, ventilation into the interior spaces. It reduces the need for artificial lighting during the day and improves indoor comfort.



There are various types of sky light. They are:

i. Fixed skylight:
   It cannot be opened and is used only for a daylight.

ii. Ventilating skylight:
   It can be opened for ventilation:

iii. Tubular skylight:
   It uses a reflective tube to bring daylight into small spaces.

iv. Dome skylight:
   It is made up of acrylic or polycarbonate with a dome shape.

3. **Compare between the stone masonry and brick masonry.**

Answer:
Stone and brick masonry are very commonly used in Nepal, especially for the residential building construction. The differences between stone and brick masonry are as follows:

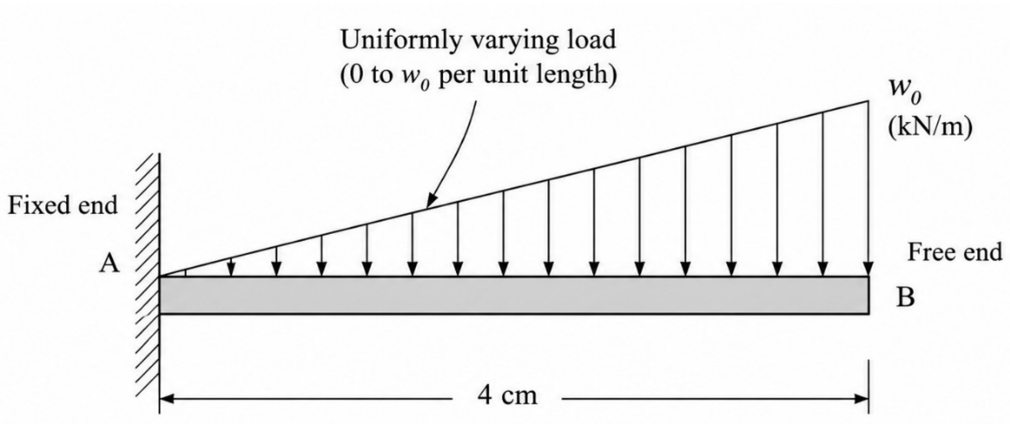
Stone masonry	Brick masonry
It is made up of natural stones such as granite, sandstone and limestone.	It is made up of burnt clay bricks or concrete bricks.
Stone masonry has higher compressive strength.	Brick masonry has lower compressive strength than stone masonry.
It is very durable and long-lasting.	It is durable but generally less durable than stone.
the masonry is heavy due to the weight of stones.	The masonry is comparatively lighter.
It requires skilled labor for dressing and laying stones.	It is generally easier and faster to construct.
Stine masonry is suitable where quality stone is locally available.	Brick masonry are widely available in most regions.
Stone has good fire resistance properties.	Bricks has excellent fire resisting properties.
It is used in foundation, retaining walls, bridges and heavy structures.	It is used in residential, commercial, institutional and partition walls.


4. **Draw opinions on the merits and demerits and B.M. diagram for the cantilever of 4 cm carrying uniformly varying load. Suggest way to enhance.**

Given,

Length of cantilever, L = 4 m

Uniformly varying load increases from 0 at the free end to w kN/m at the fixed end**.**



Shear force varies parabolically.

Maximum shear occurs at the fixed end.

Maximum shear force:

$V_{max=}wL\ /2$

For L=4 m,

$V_{max}=2w$

Bending moment varies as a cubic curve.

Maximum negative moment occurs at the fixed support.

Maximum bending moment:

$M_{max}=wL^2\ /6$

For L=4 m,

$M_{max}=w\ (4)^2/6=8w/3$

Merits of cantilever beam:

i. It provides unobstructed space below the beam.
ii. It is suitable for design of balconies and projections.
iii. Cantilever beam also provides easy success for maintenance underneath.
iv. It is economical for shorter spans.
v. Cantilever beam is a simple structural system with only one fixed support.

Demerits of a cantilever beam:

i. Larger bending moment develops at the fixed end of cantilever beam.
ii. Greater deflection can be noticed in cantilever beam, than simply supported beam.
iii. It is not economical for long spans.
iv. The failure of the fixed support can lead to collapse.

The performance can be enhanced by:

i. The depth of beam near the fixed support can be increased.
ii. Provision of adequate tensile reinforcement at the top of the beam.
iii. Using high-strength concrete and steel
iv. Using prestressed concrete for longer cantilever spans.
v. Ensuring proper anchorage and reinforcement detailing at the fixed end.

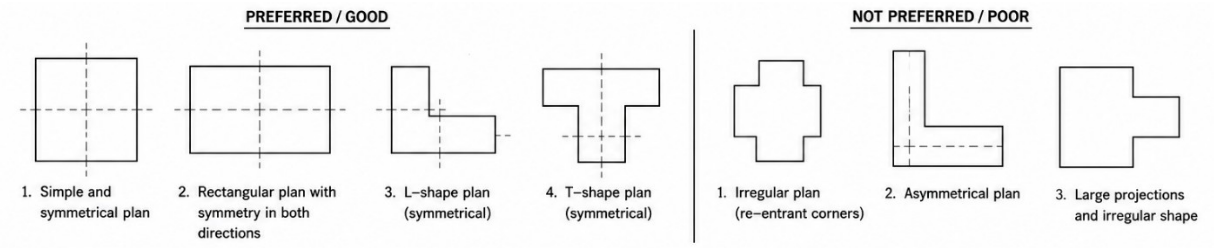
5. **Describe with sketches general, the requirements for earthquake resistant construction in terms of a. shape and proportion of plan b. size and placing of opening in wall**

Answer:

Earthquake-resistant buildings should be simple, regular, symmetrical, and well-balanced so that earthquake forces are distributed uniformly.

a. Shape:

- Buildings with simple geometry and symmetry in plan and elevation are good from earthquake resistant point of view.
- Buildings with reentrant corners like U, L, V, H suffers from more damage in earthquake.

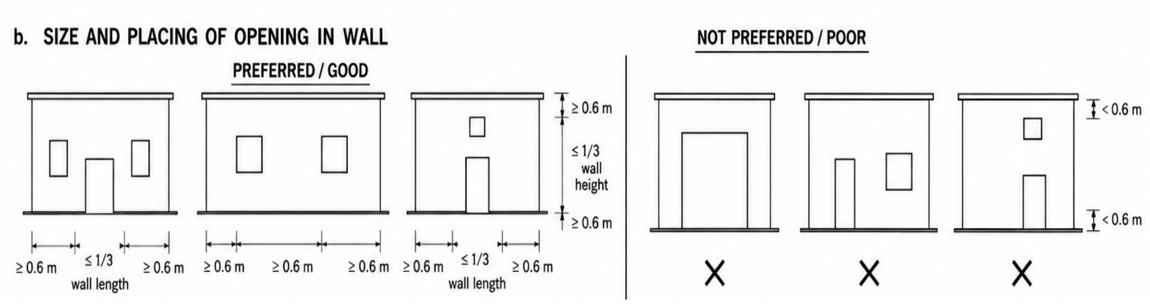


b. Proportion:

- Breadth to length ratio of buildings should be within 1:3.
- Also, breadth to length ratio for any room or area enclosed by load bearing walls inside the building shouldn't exceed 1:3.
- If the building has large horizontal dimension, then the differential arrival of seismic waves in different part of building might pose the problems.
- If the building is slender then it leads to high compressive event, resulting in rapid strength and stiffness degradation in the column.
- Height of the building shouldn't be more than 3 times the width of building.

c. Size of openings in the wall:

- Large openings in load bearing wall reduces its strength.
- Hence, the total area of openings in the wall must be less than about one-third of the total area of wall.



d. Alignment of openings:

- Doors and windows on the different floors should be placed one above another.
- This ensures continuous load transfer.

e. Number of openings:

- Too many openings reduce the wall stiffness.
- Hence, adequate wall area between the adjacent openings must be maintained.

6. **Mention with examples hierarchy of urban settlements in Nepal.**

Answer:

A permanently inhabited area with a high concentration of population, where most people are engaged in non-agricultural activities such as trade, industry, commerce, administration, education, and services can be said as urban settlements. In Nepal, urban settlements are organized in a hierarchy based on population size, administrative importance, economic functions, and the level of services provided**.**

Hierarchy of urban settlements in Nepal:

a. Metropolitan city:
   - Mahanagarpalika is the highest-level urban administrative municipality.
   - Under the Local Government Operation Act, it requires a minimum population of 500,000, an annual internal revenue of at least 1 billion NPR, and advanced infrastructure like specialized hospitals, higher education, and structured transport.
   - There are six metropolitan cities in Nepal. They are: Kathmandu Meteropolitan City, Lalitpur Metropolitan city, Bharatpur Meteropolitan city, Pokhara Metropolitan city, Birgunj metropolitan city, Biratnagar metropolitan city.

b. Sub-metropolitan city:
   - A sub-metropolitan city (Upmahanagarpalika) is an urban municipality that has a population of over 100,000 people and an annual internal revenue collection of at least 100 million Nepalese Rupees, along with necessary electricity, drinking water, communication, and paved road infrastructures.
   - They provide higher level services to surrounding municipalities and rural areas.
   - There are 11 sub-metropolitan cities in Nepal.

- Example: Ghorahi Municipality, Dang (largest sub-metropolitan city of Nepal), Nepalgunj sub metropolitan city, Banke (smallest sub metropolitan city of Nepal) and so on.

c. Municipality:
   - A municipality should have at least 9 wards and maximum 35 wards.
   - Under local governance Operation Act, it requires a minimum population of 10,000 in the districts from mountainous region, at least 40,000 in the districts from hilly region, at least 50,000 in districts from inner terai region, 75,000 in districts from terai region and 100,000 in the districts from Kathmandu Valley.
   - There are 276 municipalities in Nepal. Examples: Sheet Ganga Municipality, Arghakhanchi (Largest municipality of Nepal), Bhaktapur municipality (smallest municipality of Nepal) and so on.
   - There are 753 local levels in Nepal.

d. Small towns/ emerging urban centers:
   - Smaller urban settlements that are developing into municipalities.
   - They function as local market and service centers for nearby rural areas.
   - Examples: Phidim, Musikot, Chainpur and so on.

7. **Apartment housings are drawing attention in Nepal, describe your opinion on the merit and demerits, also illustrate the norms for apartments in inside and outside of ring roads in Nepal.**

Answer:

An apartment is a multi-storey residential building divided into separate units, where common facilities such as staircases, lifts, parking, gardens, and open spaces are shared by all residents. Rapid urbanization, population growth, limited land availability, and increasing land prices have made apartment housing increasingly popular in Nepal, especially in the Kathmandu Valley. Some of the popular apartments in Kathmandu Valley are Suncity apartments, Grande Towers, Central City apartments and so on.

Merits of apartment housing:

a. It accommodates more families on limited land area.
b. The infrastructures in apartment housing have better infrastructure such as water supply, electricity and parking.
c. The gated community with security guards and CCTV provides enhances security.
d. Modern apartments are generally designed according to Nepal Building Code (NBC) and seismic design standards.
e. It supports the compact and organized growth of city.

Demerits of apartment housing:

a. The purchase price of apartment is high and so does its maintenance costs.
b. Limited privacy due to shared walls and facilities
c. Difficulties in evacuation and management during emergencies.
d. Less open space for individual families.
e. Parking shortages and noise from neighboring apartments.

The norms for apartments inside and outside ring road are tabulated below:

Parameter	Inside Ring Road	Outside Ring Road
Maximum Ground Coverage	50%	50%
Floor Area Ratio (FAR)	3.0	3.5
Minimum Open Land Surface	20% of plot area	20% of plot area
Other Open Area	30%	30%
Front Setback	6 m	8 m

Parameter	Inside Ring Road	Outside Ring Road
Side & Rear Setback	4 m	6 m
Distance Between Two Blocks	6 m	6 m

8. **Why is it necessary to involve private sector in housing and development project of Nepal? Suggest ways of enhancing public private partnership in housing and urban development of Nepal.**

Answer:

The rapid population growth, urbanization, and migration has created a huge demand for housing and urban infrastructure in city areas of Nepal. The government alone cannot meet this demand due to financial, technical, and managerial limitations. Therefore, the private sector plays a vital role in housing and urban development through investment, innovation, and efficient project implementation.

It is necessary to involve private sector in housing and development project of Nepal because:

a. Mobilization of private investments supplements public funds for infrastructure projects that reduces burden on the government.
b. Private companies' complete projects more efficiently due to better managements and modern construction techniques.
c. Private sector can bring new technical expertise to the projects.
d. It can create job opportunity for engineers, architects, labors, technicians and so on.
e. Private developers help to increase the supply of residential housing reducing the housing shortage.
f. Sharing responsibilities with private sector allows the government to focus on regulation, planning and social housing.

PPP is known as Public Private Partnership, a collaborative arrangement where the government and private sector jointly finance, develop, operate, and maintain public infrastructure and housing projects while sharing risks and benefits. The ways of enhancing public private partnership in housing and urban development of Nepal are listed below:

a. Clear and transparent PPP laws, regulations and implementation guidelines should be developed and established.
b. Unnecessary administrative delays must be eliminated on project approval and land acquisition procedures.
c. Viability Gap Fund such as tax exemptions, low-interest loans, incentives for affordable housing projects and many such schemes can be provided.
d. Government can also provide public land or facilitate land readjustment schemes.
e. Competitive bidding and transparent contract management.
f. Encouragement to the developers to allocate a portion of projects to low- and middle-income households.
g. Training can be provided to the government officials in PPP planning, contract management and project monitoring.

9. **What are the things to keep in mind while designing and building a house? Describe briefly about each other's?**

Answer:

The design and construction of a house should ensure safety, comfort, durability, economy, and functionality. The following points should be kept in mind while designing and building a house:

a. Site selection:
   - Land with the good bearing capacity is selected.
   - Flood-prone, landslide-prone or waterlogged areas must be avoided for the construction of buildings.
   - Proper access to the facilities such as roads, water supply, electricity and drainage must be ensured.

b. Orientation of building:
   - Building is oriented to maximize the natural light and ventilation.

- Local climate and prevailing wind direction must be considered before the construction of buildings.

c. Planning and layout:
   - Provision of functional room arrangement.
   - Planning and layout must be done in such a way that privacy and convenient air and light circulation is ensured.
   - Living, kitchen, sleeping, sanitary areas must be approximately separated.

d. Structural safety:
   - Design of the structure must strictly follow the code prevalent in Nepal. i.e. Nepal Building Code.
   - Adequate foundation, beams, columns and slab must be provided.
   - Earthquake-resistant detailing must be ensured.

e. Foundation:
   - A suitable type of foundation is chosen based on the soil conditions.
   - Foundation should safely transfer loads to the ground and prevent settlement.

f. Ventilation and lightning:
   - Sufficient windows and ventilators must design and provided in the building.
   - Cross-ventilation must be ensured.
   - Use of natural daylight must be maximized.

g. Sanitation and drainage:
   - Safe disposal of wastewater should be ensured.
   - Dampness is prevented through effective site drainage.
   - Proper toilets, bathrooms and kitchen drainage must be provided.

h. Water supply:
   - Adequate and reliable source of potable water must be ensured.
   - Storage tanks and rainwater harvesting must be incorporated whenever possible.

i. Fire safety:
   - Adequate exits and escape routes must be provided.
   - Fire-resistant materials installation, fire extinguishers installation.

10. **Which building has been designed by LOUIS KHAN in Kathmandu? What are the main features of that building?**

Answer:

The famous American architect Louis I. Kahn designed the Family Planning and Maternal Child Welfare Centre**,** which is now used as the Ministry of Health and Population Building at Ramshah Path, Kathmandu. It was funded by USAID and was completed in 1960s. It was planned as part of a larger government complex, but only one block of the original master plan was completed. The building demonstrates the effective use of geometry, natural light, exposed brick, and climatic design. It is considered an important example of modern architecture in Nepal**.**

The main features of this building are listed below:

a. The building has simple, bold and symmetrical geometric design.
b. The exterior walls are made up of exposed brick which provides aesthetics as well as is durable and low maintenance.
c. Kahn emphasized the use of natural light in the building design while reducing heat gain.
d. Tall and narrow windows are provided for adequate lightning and ventilation.
e. The building has balanced and symmetrical layout providing structural and visual harmony.
f. It also includes an innovative waffle ceiling system.
g. Although modern in style, the design is responsive to Nepal’s climate, materials and cultural setting.
h. It reflects Kahn’s philosophy of combining functionality with timeless architectural expression.